YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 4 OF 10 · 40 MINUTES

Measure it well 学会准确测量

Measurements always vary a little, so scientists repeat them, work out a mean and look carefully at any result that does not fit.

- I can record repeat measurements with the correct units and calculate a mean.
- I can spot an anomaly and explain why I leave it out of the mean.

measure 测量 · **unit** 单位 · **second (s)** 秒 · **repeat** 重复 · **mean** 平均值 · **anomaly** 异常值 · **results table** 结果表格 · **reaction time** 反应时间

THE LESSON

Do Now	4	Retrieval: IV, DV and a control; why two people timed one drop differently. Today slide.
Hook	3	Teacher drops one L spinner from 2 m; three volunteers time it. Why are the times different?
Learn	7	Units, repeats, mean = total ÷ number of values, anomaly; worked example; students try Core C.
Do	18	Groups time 3 repeats per wing length (9 drops), circle anomalies with reasons, find means.
Think and Reflect	8	'Include it or leave it out?' pair talk, share with a frame; exit ticket; logbook line.

Hands-on: Groups drop their Week 3 S, M and L spinners from the 2 m (or 1.8 m) mark, 3 repeats each, stopwatch timed.

Collect or assess: Group results table with means and circled anomalies (kept for Week 5); exit ticket: circle anomaly, mean without it.

SAFETY

- Falls: never stand on chairs or tables; Dropper on the floor (1.8 m mark if needed).
- Tripping: keep bags off the floor under the drop marks.
- Scissors: carry points down. Paper clips: never unbend or point at anyone.

WATCH OUT

- Check every group divides by 2, not 3, after leaving out an anomaly; circle anomalies, never rub out.
- Short of time: same Measurer, 9 drops in about 8 min; else two repeats or the sample data card.

Language: Band 1 beside a Band 3 buddy; Chinese welcome in pair talk. Traps: mean = 平均值 (not unkind), second = 秒, Repeat 1 = 第1次.

Next week: Week 5 graphs these means and has a 10-minute checkpoint: keep the envelopes and the 6 sample data cards.

PREP FOR ONE CLASS

before the lesson about 40 min (28 could be done by someone else); after about 18 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing: worksheets x24 (double-sided, collated), kit card 5 min · reprographics

THE WEEK BEFORE

- [ ] Print 3 spare spinner sheets, 160 gsm card at 100% (unless done in Week 3) 3 min · reprographics
- [ ] Print 6 sample data cards (A5 landscape, single-sided) 2 min · technician
- [ ] Laminate the 6 sample data cards in A5 pouches (optional) optional 5 min · technician
- [ ] Check envelope spinners and wall marks (2.0 m); rebuild or re-press 13 min · technician

THE DAY BEFORE

- [ ] Collect and check 7 stopwatches and tray, scissors, clips, metre stick 5 min · technician
- [ ] Test-drop the demo L spinner from 2 m; rebuild if bent 2 min · teacher
- [ ] Plan roles (new Measurer, Dropper) and Band 1 / Band 3 buddy pairs 5 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out kit on tables and teacher desk; clear bags from under drop marks 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: spinners and results table in envelope, kit counted back 3 min · students
- [ ] Store envelopes, check results tables, note unusable data; keep cards 5 min · teacher
- [ ] Read exit tickets; note who needs a recap before the Week 5 checkpoint 10 min · teacher
- [ ] Record freezer decision; Week 6: canteen ice (Plan A) or thermometers (B) teacher

PRINT

- Worksheet x24 (5 pages, 3 sheets each, double-sided)
- Sample data cards x6 (A5 landscape, single-sided)
- Spare spinner templates x3 (A4 160 gsm card, 100% scale)
- Teacher copy: kit card (A5)

GET OR MAKE (NEW THIS LESSON)

- Stopwatches x7 (1 per group, 1 extra for Hook) and tray
- Scissors x6 pairs (repairs only)
- Paper clips, 28 mm x12 (spares)
- Metre stick x1
- A5 laminating pouches x6 (optional)

Have ready from earlier weeks: Spinner envelopes x6 (18 spinners), Demo L spinner (Week 3), Wall drop marks, 6 stations, Masking tape roll (Week 3), Role cards and frame cards, 6 sets each, Logbooks x24 in the class box, Science dictionaries x24, Pencils x24